

SanketSetu analysis report

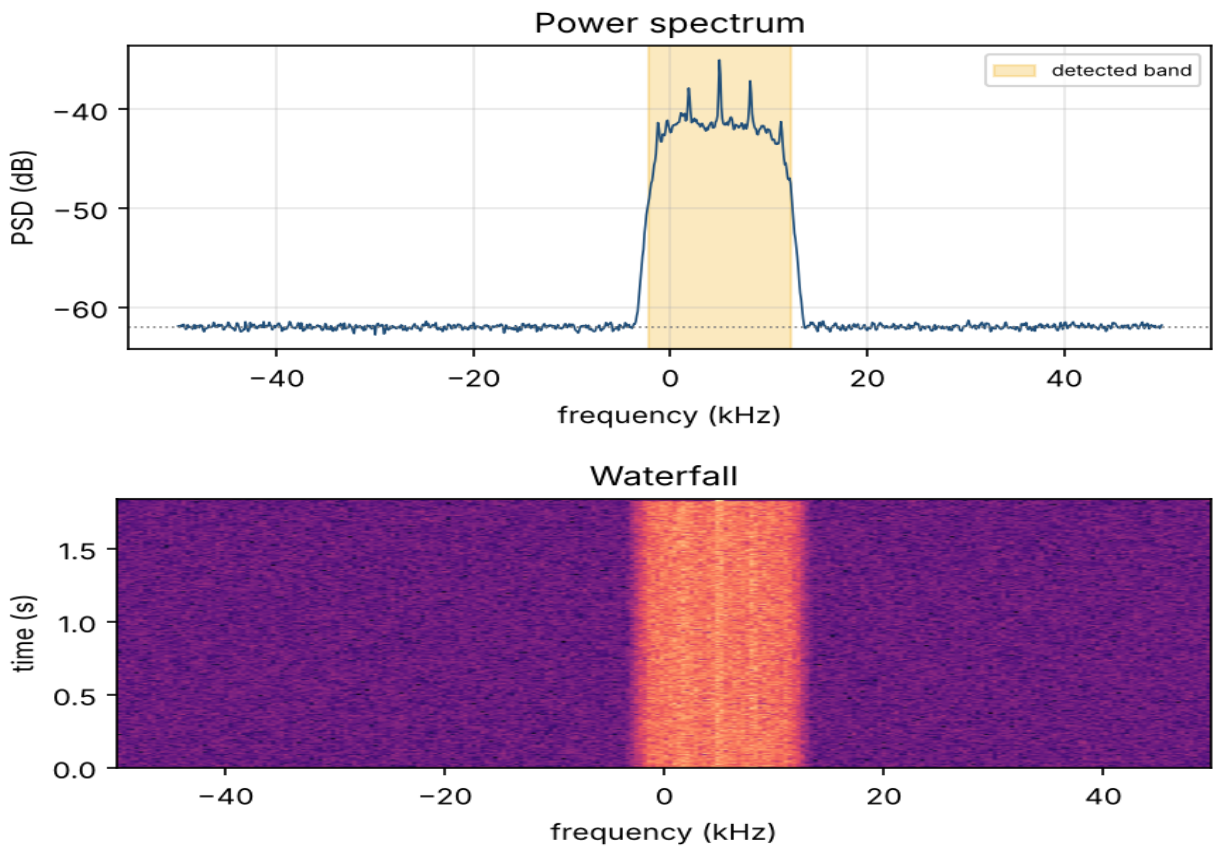
Generated 2026-10-04 20:26 | SanketSetu 0.4.0

1. Capture

File	synthetic QPSK
Kind	synthetic
Sampling rate	100,000 Hz
Samples	184,320 (1.843 s)
Notes	Generated on the server

2. Signal parameters (measured)

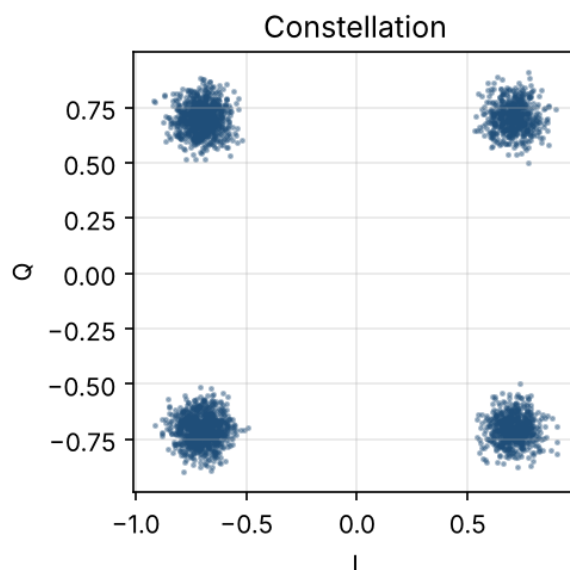
Centre frequency	4,889.8 Hz
Occupied bandwidth (99 %)	14,355.5 Hz
SNR, full band	12.0 dB
SNR, in band	20.4 dB
Noise floor	-62.0 dB
Power	0.3 dBFS
DC offset	0.007



3. Modulation and demodulation

Modulation	QPSK (from: constellation fit (overruled classifier))
Symbol rate	12,500.0 Hz (8.00 samples/symbol)
Symbol SNR (from EVM)	21.0 dB

EVM	8.9 %
Timing lock	100 %
Residual carrier offset	110.17 Hz
Bits recovered	46,070 (correct up to the phase ambiguity noted below)
Feature classifier	BPSK 79 %, QPSK 20 %, 8PSK 1 %, 16QAM 0 %
CNN (RadioML)	QPSK 61 %, 32PSK 10 %, 16PSK 7 %. Informational only: this CNN was trained on simulated RadioML data and has not been shown to agree with ground truth on this tool's own signals.
Ground truth (demo only)	BER 0.0



4. De-interleaving, FEC decoding and framing

Convolutional code	not found
LDPC	ldpc_512_256 (512, 256), rate 256/512, 88 of 88 blocks decoded, 0 bit errors corrected, start offset 504, matrix built-in
Reed-Solomon	RS(204,188), t=8, 12 codewords, 0 symbols corrected, 0 uncorrectable
Frame structure	1504 bits per frame, 12 frames, sync CCSDS ASM 0x1ACFFC1D, header 32 bits, payload 1440 bits
Ground truth (demo only)	payload byte accuracy 1.0

Recovered message (2160 bytes; first 500 characters, non-printable bytes shown as '.')

```
T 0123456789 GOLF carrier nominal uplink ECHO OSCAR WHISKEY ROMEO XRAY YANKEE XRAY CHARLIE sensor
UNIFORM carrier JULIET NOVEMBER downlink GOLF CHARLIE NOVEMBER YANKEE ECHO downlink MIKE INDIA TANGO
carrier downlink downlink LIMA ALPHA nominal ZULU status ALPHA BRAVO SIERRA MIKE PAPA beacon HOTEL
TANGO LIMA KILO packet FOXTROT LIMA ECHO FOXTROT KILO ZULU UNIFORM QUEBEC GOLF nominal nominal INDIA
CHARLIE LIMA FOXTROT nominal SIERRA SIERRA KILO SIERRA sensor INDIA status ALPHA LIMA beacon
downlink ...
```

5. Warnings from the analysis

- Classifier said BPSK but the constellation fits QPSK much better (structured data can bias the classifier features); using the better fit

Method limits: modulation, symbol-rate and code identification are estimates. Accuracy figures for this tool were measured on its own simulator, not on real captures. Interleaver type and size are operator-supplied; raw IQ sampling rate comes from the operator. Treat unconfirmed results as hypotheses.